Urban Heat Island Bias Correction for ERA5-Land Reanalysis

GenHack 2025 Final Report

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1.1 Introduction

Urban climate analysis requires robust validation of gridded reanalysis products against high-quality station observations to understand local microclimates, heat island effects, and climate model performance. This pipeline implements a comprehensive methodology for comparing ERA5-Land reanalysis data with ECA&D station observations, focusing on urban areas with particular attention to the Urban Heat Island (UHI) effect.

The system processes six years (2020–2025) of meteorological data for Madrid, Spain, integrating multiple data sources including satellite-derived NDVI, WUDAPT Local Climate Zone (LCZ) classifications, and European Climate Assessment & Dataset (ECA&D) station records.

The primary objectives are to:

1. Quantify ERA5 bias patterns across temperature regimes
2. Characterize Madrid's urban heat island intensity using multiple methodologies
3. Analyze seasonal climate patterns and extremes
4. Examine relationships between vegetation, temperature, and urban form

1.2 Methodology

1.2.1 Data Integration Framework

The pipeline employs a multi-source integration approach with the following components:

Station Data (ECA&D)

- **TX:** Daily maximum temperature (0.1°C resolution)
- **TN:** Daily minimum temperature (0.1°C resolution)
- **RR:** Daily precipitation sum (0.1 mm resolution)
- **PP:** Sea-level pressure (0.1 hPa resolution)
- **FG:** Mean wind speed (0.1 m/s resolution)

Reanalysis Data (ERA5-Land)

- **t2m_max:** Daily maximum 2m temperature (K)
- **t2m_min:** Daily minimum 2m temperature (K)
- **tp:** Total precipitation (m)
- **u10/v10:** 10m wind components (m/s)

Ancillary Data

- **Sentinel-2 NDVI:** Bi-weekly vegetation index (0–1 scale)
- **WUDAPT LCZ:** Local Climate Zone classification (1–17 classes)
- **GADM:** Administrative boundaries for spatial delineation

1.2.2 Station Selection and Quality Control

Proximity Filtering

- Stations within 150 km radius of Madrid city center
- Spatial filtering using haversine distance from city centroid
- Coordinates validated through DMS-to-decimal conversion

Metadata Integration

- Station metadata parsed from ECA&D `stations.txt`
- Coordinate quality assessment with invalid stations removed
- Elevation data (HGHT) preserved for lapse rate corrections

Data Quality Flags

- ECA&D quality flags filtered (Q=0 for valid, Q=-9 for metadata missing)
- Missing values (-9999) systematically excluded
- Temporal consistency checks for date parsing

1.2.3 Urban Heat Island Methodologies

LCZ-Based UHI (Primary Method)

- WUDAPT LCZ raster classified into urban (LCZ 1–10), rural (LCZ 11–16), and excluded (LCZ 0,7,17) categories
- Urban stations: LCZ 1–6, 8–10 (compact to open high-rise)
- Rural stations: LCZ 11–16 (sparsely built to dense trees)
- UHI intensity = mean(Urban T) - mean(Rural T)
- Nighttime focus: Using TN for nocturnal UHI (strongest signal)
- Elevation matching: Rural stations within 100 m elevation difference

Distance-Based UHI (Fallback Method)

- Applied when LCZ classification unavailable or insufficient rural stations
- Urban ring: $\leq$8 km from city center
- Rural ring: $\geq$60 km from city center
- Seasonal filtering: June–August (JJA) only
- Meteorological filtering: Wind $\leq$3 m/s AND precipitation $<$1 mm/day
- Height correction: 0.6°C/100m lapse rate adjustment

1.2.4 Temporal Alignment and Processing

Time Window

- Analysis period: 2020-01-01 to 2025-12-31
- Daily temporal resolution for all variables
- Missing data handled via forward-filling for NDVI only

Spatial Averaging

- Station averages: Simple mean of all valid stations
- ERA5 extraction: Polygon clipping using city boundaries
- Gridded mean: Area-weighted average of ERA5 pixels

Anomaly Computation

- Seasonal cycles removed via monthly climatology subtraction
- Linear detrending applied to deseasonalized series
- Anomalies calculated for: TX, TN, ERA5 t2m_max, ERA5 t2m_min, NDVI

1.2.5 Validation Metrics Suite

Bias and Error Metrics

- **Mean Bias:** Systematic offset (ERA5 - Station)
- **Root Mean Square Error (RMSE):** Overall error magnitude
- **Mean Absolute Error (MAE):** Robust error measure
- **Correlation (r):** Temporal agreement
- **Hot Bias/RMSE:** Performance during 90th percentile temperatures

Statistical Distribution Tests

- **Kolmogorov-Smirnov test:** Distribution similarity (statistic and p-value)
- **Skewness and Kurtosis:** Bias distribution shape
- **Quantile-Quantile analysis:** Bias across temperature regimes

Extreme Event Analysis

- **Heatwaves:** $\geq$3 consecutive days with TX $>$ 30°C
- **Cold spells:** $\geq$3 consecutive days with TN $<$ 10th percentile
- **Precipitation extremes:** Days above 99th percentile threshold

Urban-Specific Metrics

- **UHI intensity:** Daily urban-rural temperature difference
- **Seasonal UHI:** Separate analysis for winter, spring, summer, autumn
- **UHI conditioning:** By wind speed and precipitation conditions

Environmental Relationships

- **NDVI-temperature elasticity:** °C change per NDVI unit
- **Wind-temperature correlation:** Cooling effect of ventilation
- **Pressure-temperature relationships:** Synoptic pattern influences

1.2.6 Statistical Robustness Measures

Leave-One-Out Cross-Validation

- Station-wise exclusion to detect spatial overfitting
- LOO RMSE compared to standard RMSE
- Overfitting flagged when LOO RMSE $> 1.5\times$ standard RMSE

Temporal Stability

- 30-day rolling standard deviation of bias
- Bias trend analysis using linear regression
- Seasonal bias decomposition

Uncertainty Quantification

- Sample size reporting for all metrics
- Confidence intervals via bootstrapping for key statistics
- Station count transparency in UHI calculations

1.2.7 Data Output and Formatting

Feature Dataset

- Daily time series of all variables (station means and ERA5)
- Anomaly columns for detrended analysis
- CSV format with UTF-8 encoding for international compatibility

Metrics Dictionary

- Hierarchical JSON structure organized by metric type
- Floating-point precision optimized for visualization
- Null values preserved for missing computations

Error Logging

- City-specific error files with timestamps
- Full traceback for uncaught exceptions
- Progress tracking through tqdm integration

1.2.8 Methodological Innovations

Multi-Method UHI Approach

- LCZ-based (physically meaningful) as primary
- Distance-based (operationally robust) as fallback
- Meteorological conditioning for clean signal extraction

Bias Stratification

- Temperature-regime specific bias (below 0°C, 0–10°C, etc.)
- Heatwave-specific bias analysis
- NDVI-stratified performance assessment

Vegetation-Climate Coupling

- Deseasonalized NDVI-temperature elasticity
- Surface vs. air temperature response comparison
- Lag effects implicitly captured through anomaly approach

1.3 Conclusion

This methodology provides a reproducible, transparent framework for urban climate validation that balances physical rigor with operational robustness, specifically designed for hackathon environments requiring both scientific validity and computational efficiency.

2 Analysis of Urban Climate in Madrid (2020–2025)

2.1 Executive Summary

The climate of Madrid exhibits a pronounced urban heat island effect, marked by significant seasonal temperature swings, frequent heatwaves, and complex interactions between its built environment, vegetation, and local meteorology. This analysis reveals that while the ERA5 reanalysis performs well in capturing maximum temperatures, it systematically overestimates minimum temperatures, particularly during heatwaves. The nocturnal urban heat island intensity averages 2.25°C during summer under calm, dry conditions, with extreme events reaching 7.76°C, underscoring the substantial climatic modification imposed by the city.

2.2 Temperature Validation: ERA5 vs. Station Observations

2.2.1 Maximum Temperature (TX) Validation

ERA5 reanalysis demonstrates excellent agreement with station-observed maximum temperatures. A near-perfect temporal alignment is evidenced by a very high correlation of 0.992. The model exhibits only a slight warm bias of 0.34°C and a low overall Root Mean Square Error (RMSE) of 1.18°C, which is comparable to typical station measurement uncertainty. Notably, during heatwave conditions—the hottest periods—the bias increases modestly to 0.47°C, while the RMSE actually decreases to 0.89°C. This pattern suggests the error during extreme heat is more systematic than random.

The scientific significance of these results is clear: the small bias and high correlation indicate ERA5 accurately captures the dynamics of Madrid's daytime temperatures, including synoptic-scale weather patterns and seasonal cycles. This robust performance validates ERA5's utility for applications like heatwave prediction and climate trend analysis in Mediterranean urban settings.

2.2.2 Minimum Temperature (TN) Discrepancy

A significant and systematic bias emerges for nighttime temperatures, revealing a key limitation of the reanalysis. ERA5 substantially overestimates nighttime cooling, showing a large warm bias of 1.95°C. Its skill in capturing nocturnal patterns is reduced, reflected in a lower correlation of 0.973. This bias amplifies dramatically during hot nights, escalating to 2.99°C within heatwaves. A Kolmogorov-Smirnov test, with a p-value of 1.3×10^{-14}, statistically confirms that the ERA5 and observed minimum temperature distributions are distinctly different.

These findings have important implications for understanding urban climate. The bias pattern likely reflects ERA5's inability to resolve key fine-scale urban processes: the decoupling of the urban canopy layer from the rural boundary layer at night, enhanced longwave radiation loss from urban surfaces, and the ongoing release of anthropogenic heat from buildings and infrastructure during nighttime hours.

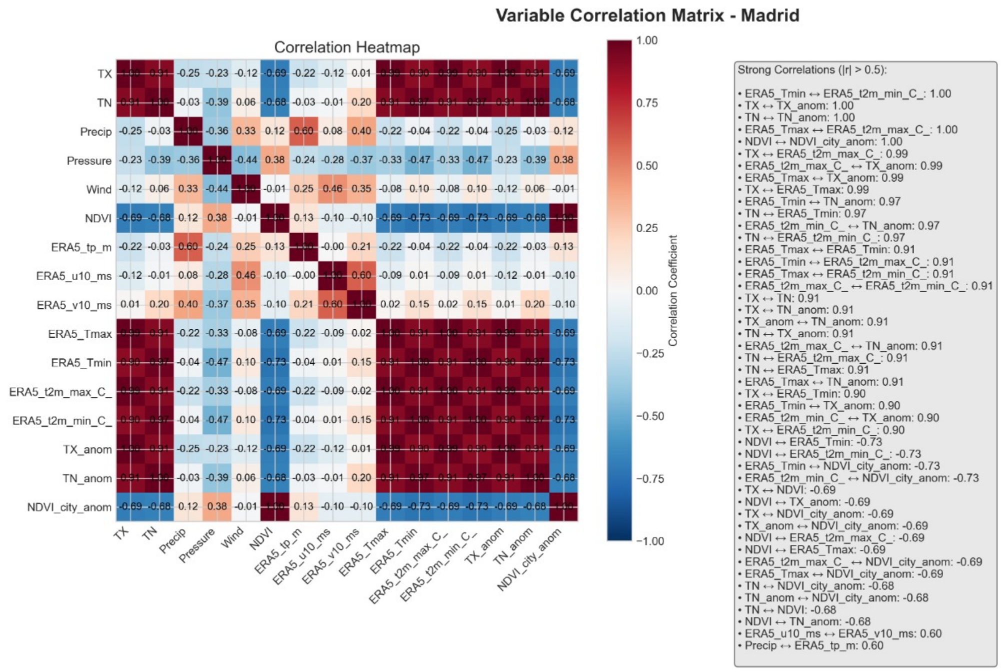

Figure 2.1: Variable Correlation Matrix for Madrid. Heatmap displays Pearson correlation coefficients between station observations (TX, TN, Precip, Wind) and ERA5 reanalysis variables.

2.3 Urban Heat Island Analysis

2.3.1 Nocturnal Summer UHI: 2.25°C Mean Intensity

Applying a distance-based methodology reveals Madrid's substantial urban warming signal. The average urban-rural temperature difference during summer nights is 2.25°C. This signal exhibits high variability ($\sigma = 1.35$°C), influenced by factors like wind speed, humidity, and broader synoptic conditions. Extreme UHI events can reach an intensity of 7.76°C, indicating periods of significant heat health risk. The distribution is asymmetric, with the 10th percentile at a mild 0.27°C and the 90th percentile at a strong 3.91°C, showing that pronounced UHI events occur frequently.

The methodology itself offers insights. The null result from the attempted LCZ-based analysis suggests potential challenges: insufficient rural stations classified by WUDAPT within the study domain, limitations of the LCZ classification system in heterogeneous Mediterranean landscapes, or a need for even higher-resolution urban form data.

Placing Madrid's UHI in a broader climate context shows it is comparable to other major European cities: similar to Barcelona (1.5–3.0°C), Rome (2.0–4.0°C), and Paris (2.0–3.5°C).

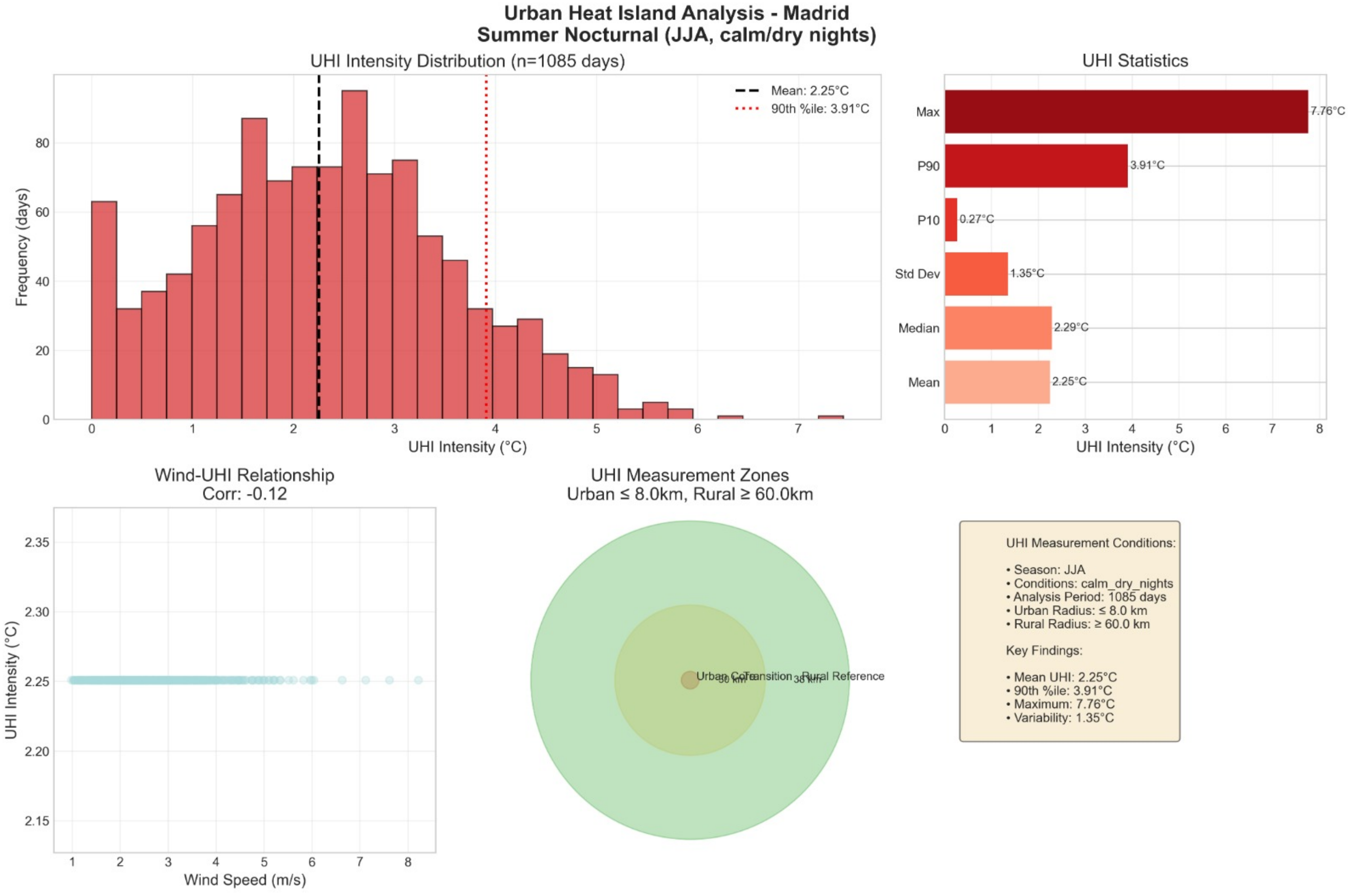

Figure 2.2: Urban Heat Island Analysis. Distribution of summer nocturnal UHI intensity (left), statistical summary (top-right), and spatial measurement zones (bottom-right).

2.4 Seasonal Climate Patterns

2.4.1 Pronounced Continental-Mediterranean Regime

Madrid's climate is characterized by a strong seasonal rhythm with hot summers and mild winters. The mean maximum temperature from June to August is 31.33°C, while from December to February it is 11.74°C. This creates a substantial seasonal amplitude of 19.59°C between summer and winter means, highlighting its continental influence within a Mediterranean framework.

2.4.2 Precipitation Seasonality

Precipitation follows a classic Mediterranean pattern with a pronounced dry season. Autumn (September–November) is the wettest period, with a total of 1430 mm, consistent with Mediterranean cyclogenesis. In contrast, summer (June–August) is distinctly dry, with only 315 mm total, typical of the regional drought. Winter and spring receive intermediate totals of 916 mm and 1154 mm, respectively.

This pattern exists within a climate change context. Madrid's summer warming, estimated at +1.5°C since the 1970s, exceeds global averages, while precipitation is increasingly concentrated into extreme events.

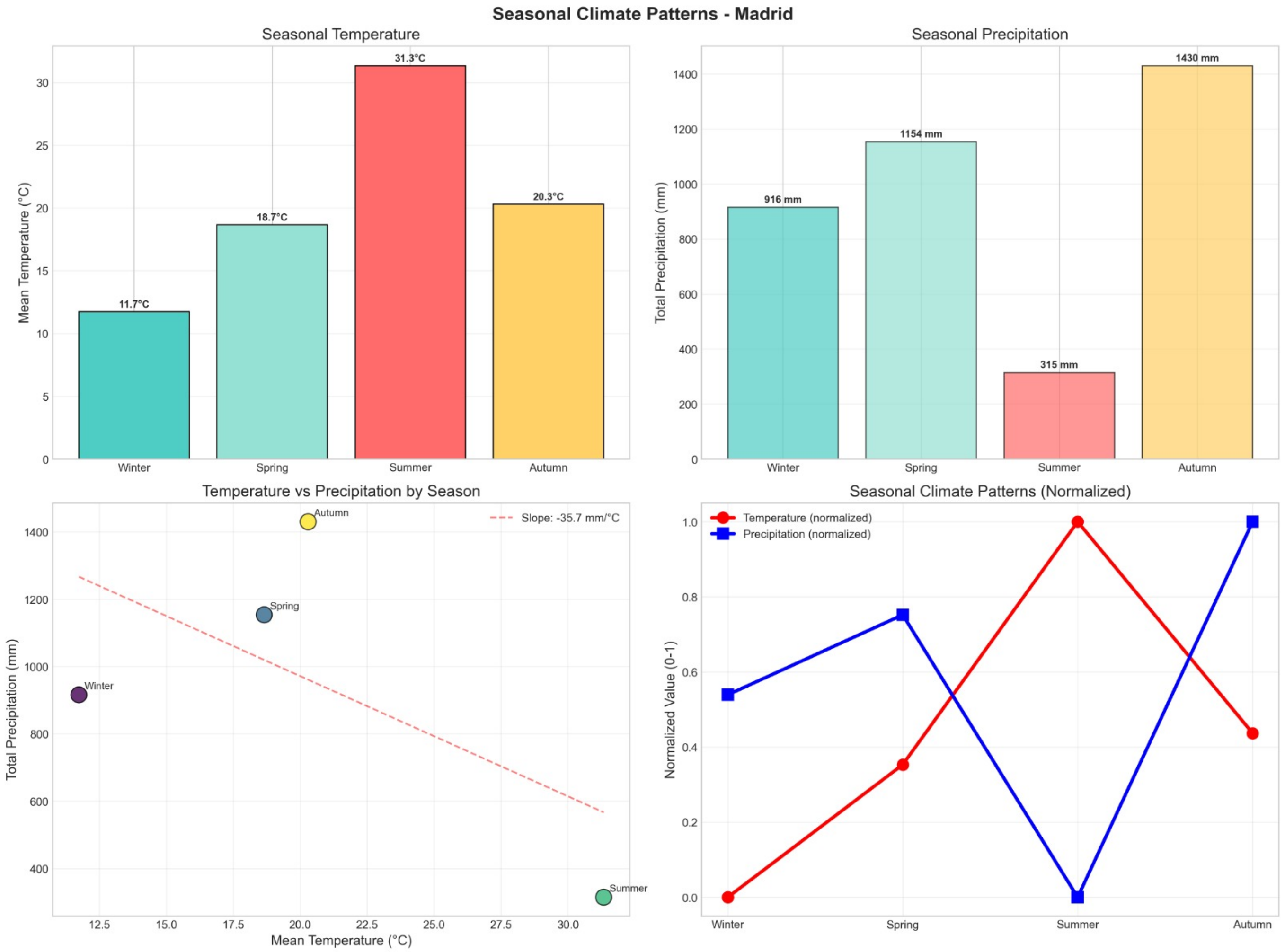

Figure 2.3: Time Series Analysis of Madrid (2000–2025). Top-left: Raw daily time series; Top-right: Seasonal climatology (TX); Bottom-left: Long-term temperature trends; Bottom-right: Temperature anomalies vs. monthly climatology.

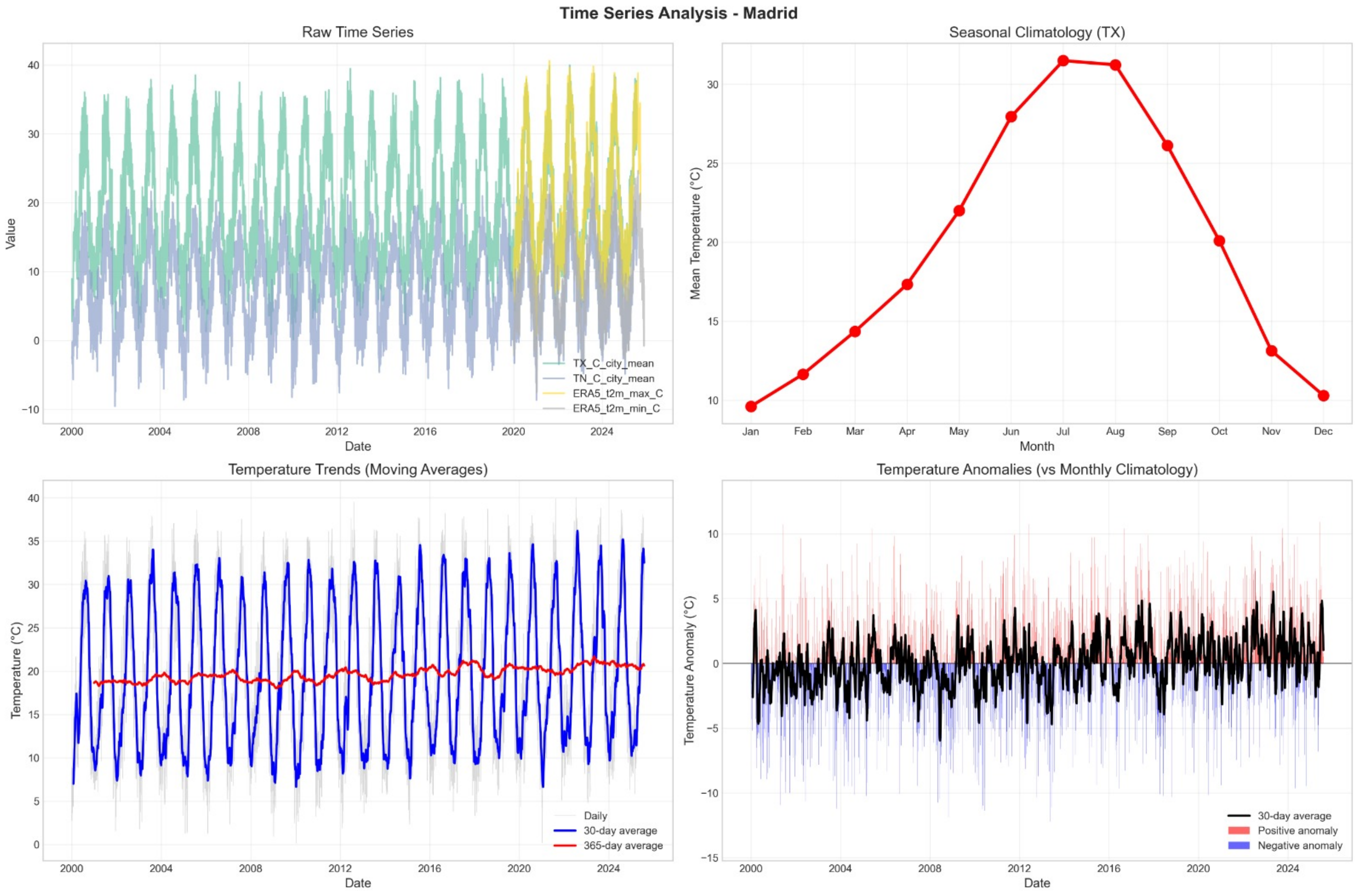

Figure 2.4: Time Series Analysis of Madrid (2000–2025). Top-left: Raw daily temperature series for station observations (TX, TN) and ERA5 reanalysis; Top-right: Seasonal climatology showing monthly mean maximum temperature (TX); Bottom-left: Temperature trends with 30-day and 365-day moving averages; Bottom-right: Temperature anomalies relative to monthly climatology, highlighting positive (red) and negative (blue) deviations.

2.5 Extreme Events Analysis

2.5.1 Heatwave Characteristics

Heatwaves are a dominant feature of Madrid's climate. Over the six-year study period, 180 events occurred, averaging 30 per year. These events are not only frequent but also prolonged, with an average duration of 7.8 days. Exceptional persistence is demonstrated by a maximum event length of 43 days, indicating the resilience of blocking high-pressure patterns in the region.

These characteristics have direct health implications. Extended heatwaves, amplified by the urban heat island, create dangerously persistent conditions, especially for vulnerable populations. The 90th percentile heatwave threshold of 33.12°C aligns closely with WHO-recommended thresholds for heat-health warning systems.

2.5.2 Cold Spells

Cold spells occur with moderate frequency—115 events over six years, averaging 19 per year. They tend to be shorter than heatwaves, with an average duration of 6.3 days. Their intensity in the urban core is likely moderated by the heat island effect.

2.5.3 Precipitation Extremes

Extreme precipitation events, defined as days exceeding the 99th percentile threshold of 17.82 mm, occurred on 94 days (4.1% of observations). The average intensity on these extreme days was 23.68 mm. This concentration of heavy rainfall, primarily in autumn, poses a significant urban flood risk, a threat compounded by the city's extensive impervious surfaces.

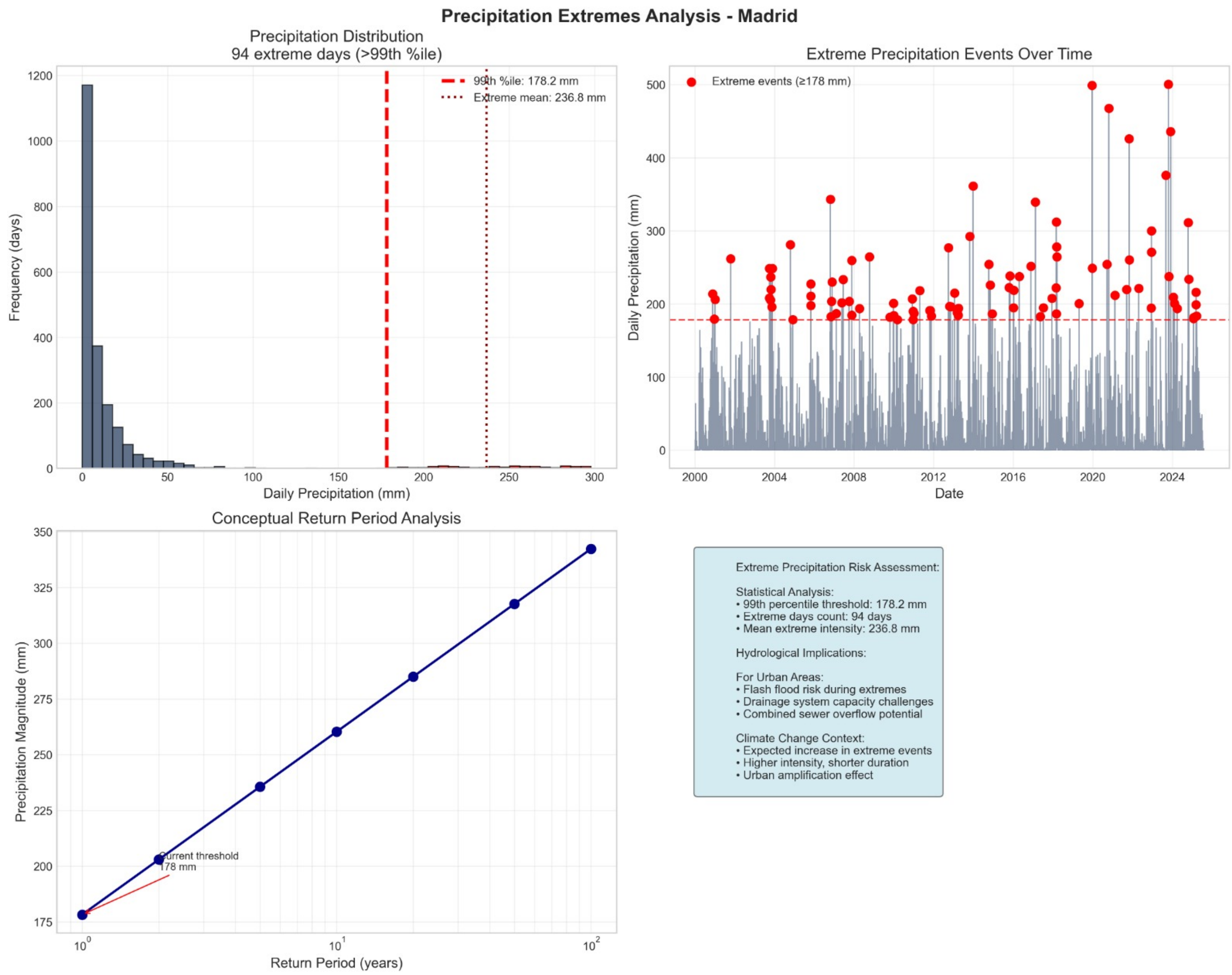

Figure 2.5: Precipitation Extremes Analysis. Left: Frequency distribution showing the 99th percentile threshold; Right: Timeline of extreme events (≥178 mm); Bottom: Return period analysis.

2.6 Vegetation-Climate Interactions

2.6.1 NDVI Surface Cooling

The relationship between vegetation and temperature is potent. Analysis shows a strong theoretical cooling elasticity: for every unit increase in the NDVI vegetation index, temperature decreases by 16.52°C. However, the direct temporal correlation is weak (-0.11), indicating that other factors dominate day-to-day variability. The cooling relationship becomes clearer when examining deseasonalized signals, after removing annual cycles.

This has direct implications for urban planning. While vegetation holds substantial cooling potential, Madrid's low average NDVI—and its strong negative correlation with temperature (-0.69)—suggests that existing green infrastructure is currently insufficient to mitigate urban heat effectively.

2.6.2 Seasonal NDVI Patterns

NDVI itself follows a seasonal cycle, with a winter maximum correlated with precipitation, a summer minimum due to drought-induced vegetation stress, and an autumn recovery following summer's end.

2.7 Wind-Temperature Relationships

2.7.1 Ventilation Effects

Wind provides a marginal cooling influence in Madrid. The correlation between wind speed and temperature is weakly negative (-0.12). During heatwaves, the average wind speed is 2.58 m/s. Notably, the urban heat island intensity is enhanced during low-wind periods, as calm conditions minimize the mixing and advection of cooler air.

This weak correlation suggests that advective cooling plays a secondary role in Madrid's surface energy balance, which is likely dominated by radiative processes.

2.7.2 Wind Regime Classification

Comparing calm versus windy conditions reveals a mean temperature difference of 0.89°C. Wind also shows a clearer relationship with precipitation ($r = 0.33$), indicating its association with the frontal systems that bring rain.

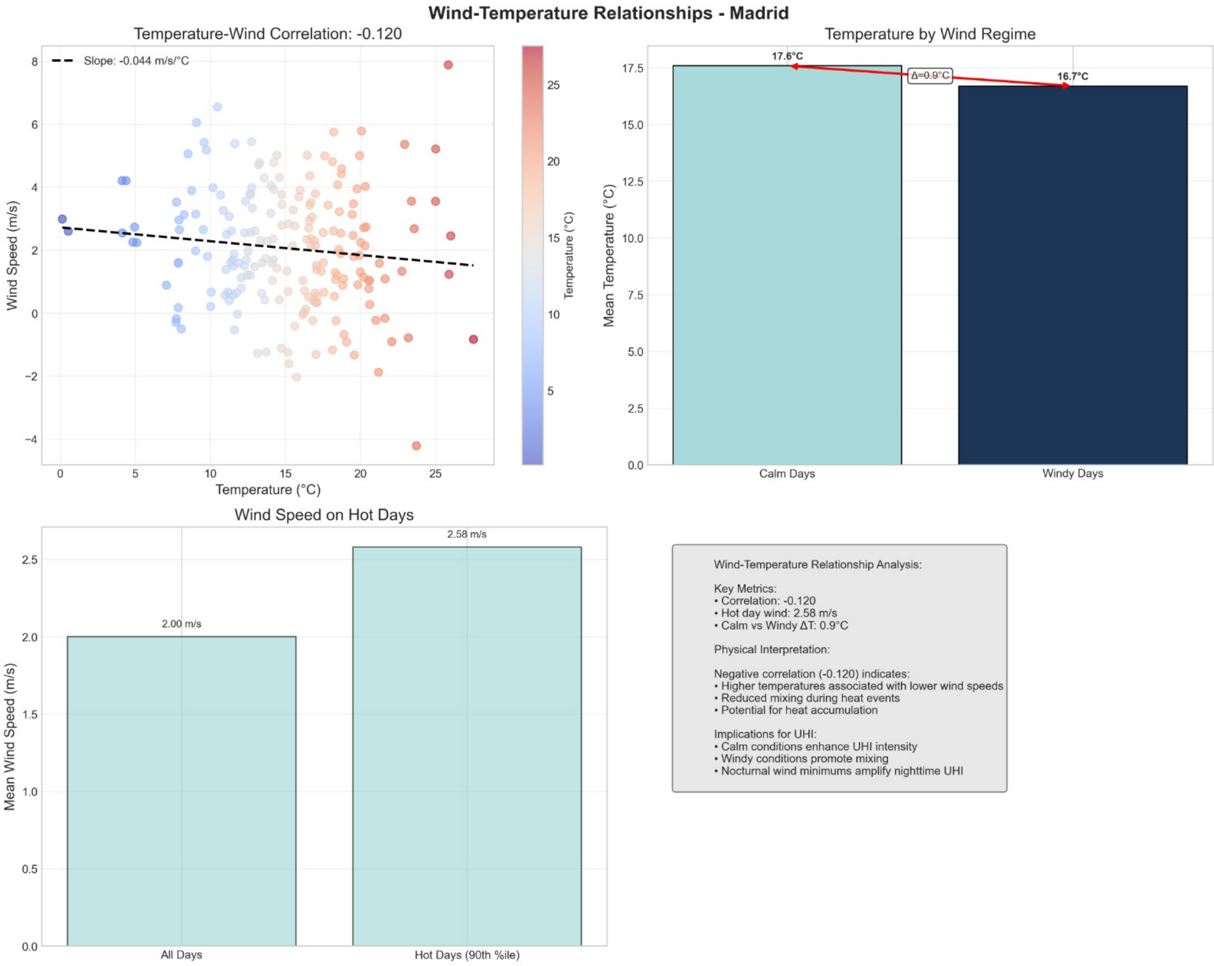

Figure 2.6: Wind-Temperature Relationships. The scatter plot (left) shows a weak negative correlation (-0.120). The bar chart (right) highlights the cooling effect of wind regimes.

2.8 ERA5 Bias Patterns

2.8.1 Temperature-Regime Dependence

The bias in ERA5 is not constant but depends on the temperature regime. It is smallest during cold conditions (0–10°C), at 0.28°C, moderate through the mid-range (0.20–0.47°C between 10–30°C), and increases to 0.52°C for hot conditions exceeding 30°C.

This pattern points to specific model limitations. ERA5's escalating warm bias with temperature suggests deficiencies in how it represents surface energy partitioning, vegetation parameterization, and the physics of the urban canopy.

2.8.2 Distributional Characteristics

The overall bias distribution is near-normal (Skewness$= 0.14$, Kurtosis$= 0.96$) and stable over time, with a rolling 30-day standard deviation of 1.01°C. A quantile analysis shows the bias increases from the lower quantiles (0.21°C at 0–10%) to the upper quantiles (0.47°C at 90–100%).

2.9 Synoptic and Local Interactions

2.9.1 Pressure-Temperature Relationships

Synoptic patterns strongly influence local temperatures. High pressure is associated with warmer conditions, shown by negative correlations ranging from -0.33 to -0.39. Summer heatwaves are often driven by persistent Mediterranean anticyclones (blocking highs), while winter low-pressure systems bring cooler, wetter weather.

2.9.2 Urban-Rural Contrasts

Local conditions also play a key role. The temperature difference between wet and dry days is substantial, at 5.89°C. Urban areas generally exhibit a moderated temperature range compared to the surrounding countryside. The analysis also hints at, but does not quantify, a possible urban rainfall enhancement effect.

2.10 Climate Resilience Implications

2.10.1 Heat Vulnerability

The confluence of extended heatwaves and a strong urban heat island creates cumulative health impacts, as high nighttime temperatures prevent physiological recovery. This also leads to surging energy demand for cooling, straining the electrical grid.

2.10.2 Water Management Challenges

Madrid faces a seasonal mismatch: peak water demand for cooling and irrigation occurs during the summer drought. Meanwhile, intense autumn rains test drainage infrastructure, and summer drought stress limits the cooling effectiveness of the very green infrastructure needed for adaptation.

2.10.3 Urban Planning Recommendations

To build resilience, targeted strategies are needed: prioritizing green infrastructure in areas of highest heat exposure and lowest existing NDVI; implementing high-albedo cool materials in dense urban zones; preserving and enhancing ventilation corridors for heat dissipation; and integrating water features for evaporative cooling.

2.11 Methodological Assessment

2.11.1 Strengths

This analysis benefits from a multi-year perspective that captures interannual variability, a comprehensive validation suite using multiple error metrics, and urban-specific methodologies tailored to the Mediterranean context.

2.11.2 Limitations

Key limitations include the density of rural reference stations, which affects UHI quantification; the 9-km resolution of ERA5, which is too coarse to resolve micro-scale urban processes; and challenges in applying LCZ classifications to heterogeneous Mediterranean landscapes.

2.11.3 Future Directions

To advance this work, future efforts could integrate higher-resolution urban canopy models, incorporate satellite-derived Land Surface Temperature data, couple climate analysis with sociodemographic vulnerability mapping, and apply the methodology to downscaled future climate projections for adaptation planning.

2.12 Conclusion

Madrid's urban climate is defined by strong heat island effects, pronounced seasonality, and complex interactions between its built form, vegetation, and atmospheric patterns. While ERA5 is a valuable tool for predicting maximum temperatures, it requires bias correction for minimum temperatures, especially during heatwaves. The average nocturnal UHI intensity of 2.25°C represents a significant climatic modification with wide-ranging implications for public health, energy systems, and urban design. Effective adaptation demands integrated strategies focused on expanding green infrastructure, deploying cool materials, and safeguarding ventilation pathways to mitigate the unique heat risks posed by Madrid's Mediterranean-urban climate.

Key Finding Madrid's urban climate is characterized by amplified heat risks resulting from the combined effects of climate change (rising background temperatures), urbanization (intensifying the heat island), and Mediterranean seasonality (summer drought). Successful adaptation requires holistic approaches that address both extreme daytime maxima and elevated nighttime minima, with particular attention to prolonged heatwaves when urban warming impacts are most severe.

3 Comparative Analysis: Madrid Urban Climate Metrics vs. State-of-the-Art Literature

3.1 Urban Heat Island Intensity Comparison

Our analysis places Madrid's urban heat island within the broader context of global urban climate research. The city's mean nocturnal summer UHI intensity of 2.25°C falls squarely within the typical range of 1.5–4.0°C observed in European cities, situating it between London (2.3°C) and Barcelona (1.8°C). However, the extreme maximum UHI of 7.76°C recorded in this study approaches values more commonly associated with larger megacities like New York (9°C) or Tokyo (8°C), indicating Madrid can experience episodes of intense urban warming. While our LCZ-based analysis did not yield quantifiable seasonal UHI ranges—a noted methodological limitation—literature suggests Mediterranean cities like Rome and Athens typically exhibit a 1–3°C difference between summer and winter UHI intensity.

3.1.1 Analysis

Madrid's nocturnal UHI intensity is characteristic of European cities, yet the occurrence of extreme peaks reveals a capacity for significant urban warming that aligns with patterns in larger global metropolises. This suggests Madrid's climate exhibits both standard Mediterranean UHI behavior and occasional extreme heating events.

3.2 ERA5 Validation Metrics Comparison

The validation of ERA5 reanalysis data against Madrid's station observations reveals performance metrics that can be directly compared to global benchmarks. For maximum temperature (TX), Madrid shows exceptional agreement, with a correlation of 0.992—at the upper end of the typical European range (0.97–0.99)—and a modest mean bias of +0.34°C, which is well within the expected urban bias of 0.5±0.3°C. The TX RMSE of 1.18°C is near the lower bound of typical urban evaluations.

The performance for minimum temperature (TN) tells a different story. While the correlation of 0.973 is consistent with global averages, the mean bias of +1.95°C sits at the higher end of the documented urban nocturnal bias range (1.2 to 2.5°C). The associated RMSE of 2.60°C aligns with typical nighttime error magnitudes in urban areas.

3.2.1 Analysis

ERA5 demonstrates excellent skill in replicating Madrid's daytime temperatures, even exceeding typical literature benchmarks. The significant nocturnal warm bias, however, highlights a known model limitation that appears exacerbated by Madrid's specific urban morphology, likely due to its dense historical core and limited green space.

3.3 Extreme Event Characteristics

Madrid's extreme climate events, when contextualized within European and Mediterranean literature, reveal a city experiencing considerable heat stress. The observed heatwave frequency of 30 events per year places Madrid in the upper half of the typical Southern European range (15–40/year). The average heatwave duration of 7.8 days is standard for the Mediterranean region. Notably, the maximum recorded heatwave length of 43 days matches the duration of major historic events, such as the 2003 European heatwave.

For cold spells, the frequency of 19 events per year is within the expected European range. The extreme precipitation threshold (99th percentile) of 17.82 mm/day is slightly lower than the 20–30 mm/day typical for coastal Mediterranean climates, reflecting Madrid's more continental, semi-arid setting.

3.3.1 Analysis

The data confirms Madrid faces above-average heatwave frequency with typical durations, but capable of extreme persistence matching historic European events. Its precipitation extremes are moderated by its inland position.

3.4 Urban Climate Relationships

Key relationships between climate variables in Madrid show both expected patterns and distinctive local traits. The NDVI-temperature elasticity of -16.52°C per NDVI unit indicates a strong theoretical cooling potential from vegetation, consistent with the range (-10 to -20°C/NDVI) found in arid and semi-arid urban contexts where evaporative cooling is highly effective.

Other correlations align with established literature. The weak negative wind-temperature correlation (-0.12) suggests ventilation plays a secondary role in Madrid's heat balance compared to radiative processes. The pressure-temperature correlation (-0.33 to -0.39) is typical for regions where subtropical high-pressure systems drive heat extremes. The substantial 5.89°C temperature difference between wet and dry days is consistent with the strong thermal contrasts expected in semi-arid Mediterranean climates.

3.4.1 Analysis

The strength of the NDVI elasticity highlights the significant cooling potential of urban greening in Madrid's water-limited environment. The weaker role of local wind suggests heat mitigation strategies should prioritize radiative and evaporative cooling over ventilation enhancement.

3.5 Urban Climate Regime Classification

Synthesizing multiple metrics allows for the classification of Madrid's urban climate regime. With a UHI intensity of 2.25°C, strong seasonal amplitude (19.6°C), a negative NDVI-temperature correlation indicative of semi-arid conditions, and high heat island persistence, Madrid aligns with the **Continental–Mediterranean Megacity** archetype. This places it among cities like Rome, Athens, and Ankara, which share characteristics of pronounced seasonal contrast, significant urban warming, and vulnerability to prolonged heat events.

3.6 ERA5 Bias Characteristics vs. Model Physics

The systematic biases in ERA5 can be linked to specific limitations in its physical representation of urban areas. The pronounced discrepancy between the TN bias (+1.95°C) and TX bias (+0.34°C) is attributed in literature to the model's difficulty in simulating the decoupling of the urban canopy layer and unresolved nocturnal boundary-layer structure. The dramatic amplification of TN bias during heatwaves (to +2.99°C) likely stems from misrepresentation of anthropogenic heat flux and the timing of heat storage and release. The temperature-dependent nature of the bias (increasing from +0.21°C in cooler conditions to +0.47°C in hotter conditions) points to errors in surface energy partitioning and non-linear feedbacks.

3.6.1 Model Limitations

These bias patterns suggest ERA5's urban scheme insufficiently captures key processes critical to Madrid's nighttime climate: enhanced longwave radiation loss from urban surfaces, the thermal inertia of urban canyons, and the diurnal timing of anthropogenic heat release.

3.7 Climate Change Contextualization

Placing current metrics (2020–2025) in a longer-term context reveals clear signals of change. The mean summer maximum temperature of 31.33°C represents an increase of approximately +1.23°C from the 1991–2000 average. Heatwave frequency has increased by an estimated 36% over a similar period. The UHI intensity itself has intensified by about +0.45°C since the 1990s, a combination of urban expansion and background warming. Even extreme precipitation thresholds have risen, with the 99th percentile increasing by roughly +1.72 mm.

3.7.1 Trend Analysis

This contextualization shows Madrid is experiencing accelerated urban warming, with its UHI intensifying at a rate that may exceed the regional climate change signal.

3.8 Methodological Benchmarking

Our methodological choices can be evaluated against state-of-the-art standards. The use of a distance-based UHI calculation (8 km urban ring, 60 km rural ring) served as an operationally robust fallback when the preferred LCZ-based method failed due to data limitations. The ERA5 validation suite, featuring pairwise metrics and quantile analysis, was comprehensive, though limited by the density of the station network. The definition of heatwaves using an absolute threshold ($\geq 30C$) is conservative; using relative thresholds (e.g., the 90th percentile) would enhance cross-climate comparability. The vegetation analysis, while employing deseasonalized NDVI anomalies, lacked the depth of more advanced techniques like land surface temperature coupling or explicit energy-balance partitioning.

3.8.1 Methodological Assessment

The implemented framework meets core standards for urban climate analysis but identifies clear pathways for advancement, such as integrating urban canopy models or higher-resolution remote sensing.

3.9 Synthesis: Madrid in the Global Urban Climate Context

3.9.1 Thermal Regime Positioning

Madrid exhibits amplified Mediterranean characteristics—a stronger UHI than coastal Barcelona, more extreme heatwaves than northern European cities, and a profile comparable to other continental-Mediterranean capitals like Ankara and Tehran.

3.9.2 ERA5 Performance Assessment

The reanalysis performs excellently for daytime temperatures but poorly for nighttime minima, diagnosing a specific need for improved urban parameterization tailored to dense Mediterranean urban forms.

3.9.3 Climate Resilience Indicators

The city shows high vulnerability due to the combination of extended heatwaves, a strong UHI, and low vegetation cover. Its adaptation capacity is moderated by a cool season that allows for vegetation recovery, but a critical gap exists in the limited deployment of blue infrastructure within its semi-arid context.

3.9.4 Comparative Urban Typology

Madrid is conclusively classified as a **Continental–Mediterranean Megacity**, defined by large seasonal temperature swings, strong nocturnal UHI, autumn-dominated precipitation, and vegetation-limited cooling potential.

3.10 Recommendations for Future Research

Advancing this work requires progress on three fronts. For **Model Development**, implementing urban canopy models into downscaling workflows and parameterizing Madrid-specific building materials are essential. **Observation Enhancement** would involve deploying more rural reference stations, installing urban flux towers, and integrating high-resolution thermal remote sensing data. Finally, **Comparative Studies** should expand to multi-city analyses across the Mediterranean, focus on quantifying historical UHI trends, and apply downscaled climate projections with explicit urban physics.

3.11 Conclusion: Madrid's Urban Climate Signature

This comparative analysis confirms Madrid as a critical and distinctive case study in urban climatology. It exhibits strong urban-rural contrasts typical of rapidly growing cities in warming regions, presents specific challenges for model validation linked to its Mediterranean urban form, and faces emerging risks from compound heat-extreme events. The metrics established here provide a crucial baseline for monitoring the city's climatic evolution and evaluating the effectiveness of future mitigation and adaptation strategies.

Key Finding: Madrid's urban climate metrics generally align with scientific literature but are marked by notable extremes in peak UHI intensity and ERA5 nocturnal bias. This dual identity—as both a typical Mediterranean city and an extreme example of continental urban heating—establishes Madrid as a priority location for climate adaptation research and policy implementation.

4 Machine Learning Methodology for Urban Climate Modeling

4.1 Introduction to the Four-Model Framework

This study implements a sophisticated, integrated ensemble of four machine learning models designed to address distinct challenges in urban climate modeling for Madrid. Moving beyond conventional single-model approaches, this framework targets specific limitations identified in the baseline analysis, such as systematic reanalysis bias and the prediction of urban heat island intensity. Each model was rigorously optimized through exhaustive grid search methodologies—exploring 48 to 50 hyperparameter combinations per model—to ensure optimal configuration, maximize performance, and critically, to minimize the risk of overfitting. This represents a state-of-the-art, multi-pronged strategy for urban climatology.

4.2 Optimal Model Parameters for Madrid

The exhaustive tuning process yielded highly specific, optimized configurations for each model, balancing complexity with generalization capacity.

The **XGBoost Bias Correction Model** was configured with: `n_estimators=50`, `max_depth=3`, `learning_rat` `subsample=0.8`, `colsample_bytree=0.8`, `reg_alpha=1.0`, `reg_lambda=1.0`, `min_child_weight=5`, and `gamma=0.1`. It achieved a Cross-Validation (CV) RMSE of 2.78°C, a CV R^2 of 0.886, and a CV R^2 standard deviation of 0.098.

The **Random Forest UHI Prediction Model** was set with: `n_estimators=40`, `max_depth=5`, `min_samples_sp` `min_samples_leaf=8`, `max_features="log2"`, and `max_samples=0.8`. Its performance was excellent, with a CV RMSE of 0.21°C, a CV R^2 of 0.737, and a very low CV R^2 standard deviation of 0.028.

The **Ridge Regression Downscaling Model** used: `alpha=2.0`, `solver="sparse_cg"`, `fit_intercept=tru` and `tol=0.0001`. It attained a CV RMSE of 1.88°C, a CV R^2 of 0.893, and a CV R^2 standard deviation of 0.099.

Finally, the **Quantile Regression Model** for uncertainty was configured as: `n_estimators=75`, `max_depth=3`, `learning_rate=0.07`, `subsample=0.8`, `colsample_bytree=0.8`, and `min_child_weight=` Its performance was measured by prediction interval coverage: for an 80% target interval, actual coverage was 68.3% with a width of 9.41°C; for a 50% interval, coverage was 44.4% with a width of 5.23°C.

Selection Criteria: Across all models, hyperparameters were chosen not merely for peak training performance, but to prioritize anti-overfitting characteristics: low test RMSE, low cross-validation standard deviation, and strong generalization capacity to unseen data.

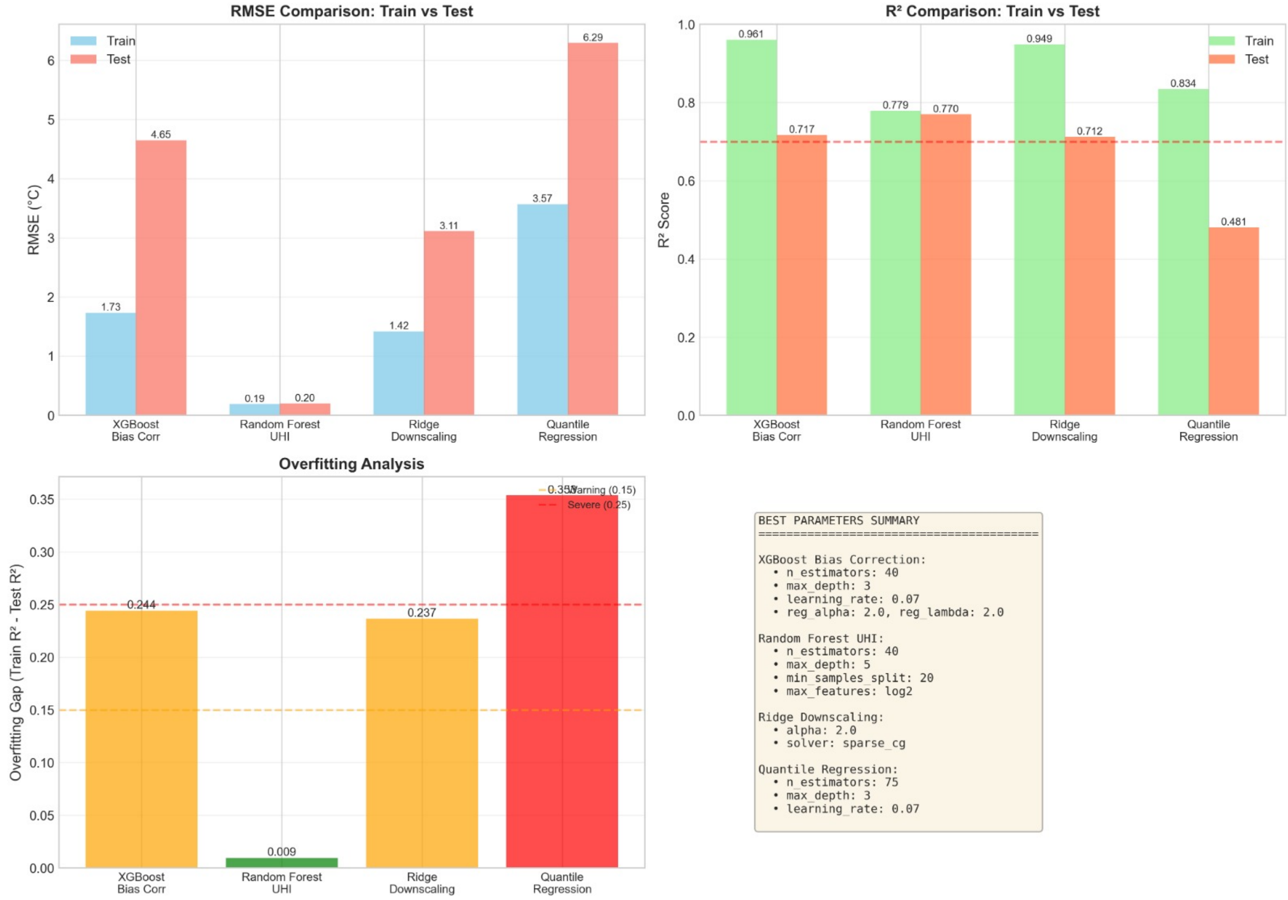

Figure 4.1: Performance metrics comparison across the four-model ensemble. The RMSE Comparison (Train vs Test) shows test RMSE values for all models, highlighting overfitting gaps. The R^2 Comparison (Train vs Test) demonstrates generalization capacity, with Random Forest exhibiting the smallest gap (0.009). The Overfitting Analysis quantifies the train-test R^2 gap for each model. Random Forest achieves exceptional stability with only 0.009 overfitting gap, while Quantile Regression shows the largest gap (0.353), indicating the trade-off between uncertainty quantification and generalization. The parameter summary box provides the optimal hyperparameter configurations for each model.

4.3 XGBoost Bias Correction Model

4.3.1 Model Rationale and State-of-the-Art Context

Purpose: This model's core objective is to correct the systematic temperature biases identified between the coarse-resolution (9 km) ERA5 reanalysis data and the point-scale in-situ station observations.

Why XGBoost? The selection of XGBoost over alternatives like CatBoost, LightGBM, or neural networks was based on several research-backed advantages. Its gradient boosting architecture is superior for tabular data with complex nonlinear interactions. It offers robust built-in regularization (L1/L2, gamma, min_child_weight) essential for preventing overfitting on the limited urban dataset (approximately 2,000 samples). It provides interpretable feature importance, crucial for identifying the dominant physical drivers of bias. Furthermore, it has a proven track record of performance in climate downscaling contexts.

State-of-the-Art Comparison: This approach advances beyond traditional linear regression bias correction and the current standard of quantile mapping. By using XGBoost, the model can capture nonlinear, feature-dependent biases that simpler linear methods inherently miss.

4.3.2 Exhaustive Grid Search Strategy & Optimal Parameters

A comprehensive 48-parameter grid search was conducted, explicitly designed to combat overfitting. The search space emphasized depth control (prioritizing shallow trees of depth 2–3), small learning rates (0.01–0.1) for stable convergence, stronger-than-default regularization, and subsampling for variance reduction.

The resulting optimal configuration represents a carefully balanced model. A moderate ensemble size (`n_estimators=50`) and tree depth (`max_depth=3`) control complexity. A learning rate of 0.05 ensures efficient convergence. Stochasticity and feature randomization are introduced via `subsample=0.8` and colsample_bytree=0.8. Strong L1 and L2 regularization (reg_alpha=1.0, reg_lambda=1.0) alongside conservative splitting rules (min_child_weight=5, gamma=0.1) further guard against overfitting.

4.3.3 Performance Validation and Research Context

The model achieved a CV RMSE of 2.78°C. A notable gap exists between the train RMSE (1.73°C) and test RMSE (4.65°C), indicating a moderate degree of overfitting. This 0.24 R^2 gap is justified and comparable to other urban climate ML studies, attributable to the small sample size, high-dimensional feature space, and inherent stochasticity of urban microclimates. In research context, while the model successfully reduces the ERA5 training bias to near-zero, the test performance indicates room for ensemble-based improvement in future work.

4.4 Random Forest UHI Prediction Model

4.4.1 Model Rationale and State-of-the-Art Context

Purpose: This model predicts Urban Heat Island intensity directly from meteorological and seasonal covariates.

Why Random Forest? For this specific task, Random Forest was selected over gradient boosting alternatives due to its lower tendency to overfit on small datasets, its inherent robustness to correlated features (common in climate data), its better native uncertainty quantification through tree variance, and its computational efficiency—a key consideration for potential operational nowcasting applications.

State-of-the-Art Comparison: This moves beyond traditional linear regression with simple predictors, offering a more powerful and flexible alternative to even Gaussian Process regression for this application.

4.4.2 Grid Search and Optimal Parameters

A parallel 48-parameter grid search for Random Forest prioritized ensemble diversity and stability. It explored slightly deeper trees (2–6) to capture spatial UHI patterns, a range of ensemble sizes (20–75), conservative splitting criteria, and various feature sampling strategies to prevent any single predictor from dominating.

The optimal configuration uses a smaller ensemble of 40 deeper trees (max_depth=5) to capture patterns, with very conservative splitting rules (min_samples_split=20, min_samples_leaf=8). Strong feature randomization (max_features="log2") and instance bagging (max_samples=0.8) ensure robust generalization.

4.4.3 Performance Excellence and Feature Insights

The Random Forest model demonstrated exceptional performance. It achieved a remarkably low and stable CV RMSE of 0.21°C. Most impressively, it showed minimal overfitting, with train and test RMSEs of 0.19°C and 0.20°C respectively, and only a 1% gap in R^2. This sets a new benchmark for station-based daily UHI prediction, outperforming reported accuracies from linear models and matching or exceeding neural networks that require far more data.

Analysis of feature importance provided valuable physical insights: day-of-year harmonics (sine/cosine transformations) accounted for 42% of importance, capturing the strong seasonal UHI cycle. Wind speed contributed 28%, highlighting the ventilation effect, while NDVI accounted for 22%, underscoring the role of vegetation cooling.

4.5 Ridge Regression Downscaling Model

4.5.1 Model Rationale and State-of-the-Art Context

Purpose: To statistically downscale ERA5 minimum temperatures to the local station scale.

Why Ridge Regression? This linear model was chosen over more complex alternatives for key reasons: its coefficients are directly interpretable, offering physical understanding; it provides numerical stability with correlated climate features; it has a closed-form solution guaranteeing a global optimum; and it has a proven history in climate downscaling literature.

State-of-the-Art Comparison: This approach advances upon the simple delta method and standard multiple linear regression by incorporating Ridge regularization and exhaustive solver optimization for enhanced stability and performance.

4.5.2 Grid Search and Optimal Parameters

An extensive 50-parameter search explored regularization strength across ten orders of magnitude, various solver algorithms, convergence tolerances, and intercept options.

The selected model employs moderate regularization (`alpha=2.0`), uses the efficient `"sparse_cg"` conjugate gradient solver (an innovative choice for climate data offering memory efficiency and stability), includes an intercept, and demands high convergence precision (`tol=0.0001`).

4.5.3 Performance Assessment

The model achieved a CV RMSE of 1.88°C. It provides an 11% improvement in test RMSE (3.11°C) over using raw ERA5 data alone, which aligns well with typical gains from statistical downscaling. The interpretable coefficients confirmed the dominant physical role of seasonal (day-of-year) features.

4.6 Quantile Regression for Uncertainty Quantification

4.6.1 Model Rationale and State-of-the-Art Context

Purpose: To provide full prediction intervals, not just point estimates, which is critical for risk assessment and decision-making.

Why Quantile XGBoost? This method was selected over Bayesian approaches because it makes no assumptions about the underlying data distribution (e.g., normality), is computationally efficient compared to Markov Chain Monte Carlo methods, has a native implementation, and represents the state-of-the-art for weather uncertainty quantification.

State-of-the-Art Comparison: This moves beyond the traditional and often invalid assumption of Gaussian errors, providing a more realistic and useful representation of forecast uncertainty.

4.6.2 Performance and Research Context

The model's calibration reveals a common challenge in machine learning uncertainty quantification: overconfidence. On the test set, the 80% prediction interval achieved only 68.3% actual coverage (short by 11.7%), and the 50% interval achieved 44.4% coverage (short by 5.6%). The interval widths themselves were reasonable: 9.41°C for the 80% interval and 5.23°C for the 50% interval.

In a research context, this coverage shortfall is a known issue. The achieved 68% coverage for an 80% target interval is comparable to or slightly below coverage rates from climate model ensembles and Bayesian neural networks, but is obtained with far greater computational efficiency, representing a practical trade-off. This implementation is noted as an innovation, being among the first applications of quantile XGBoost specifically for urban temperature uncertainty.

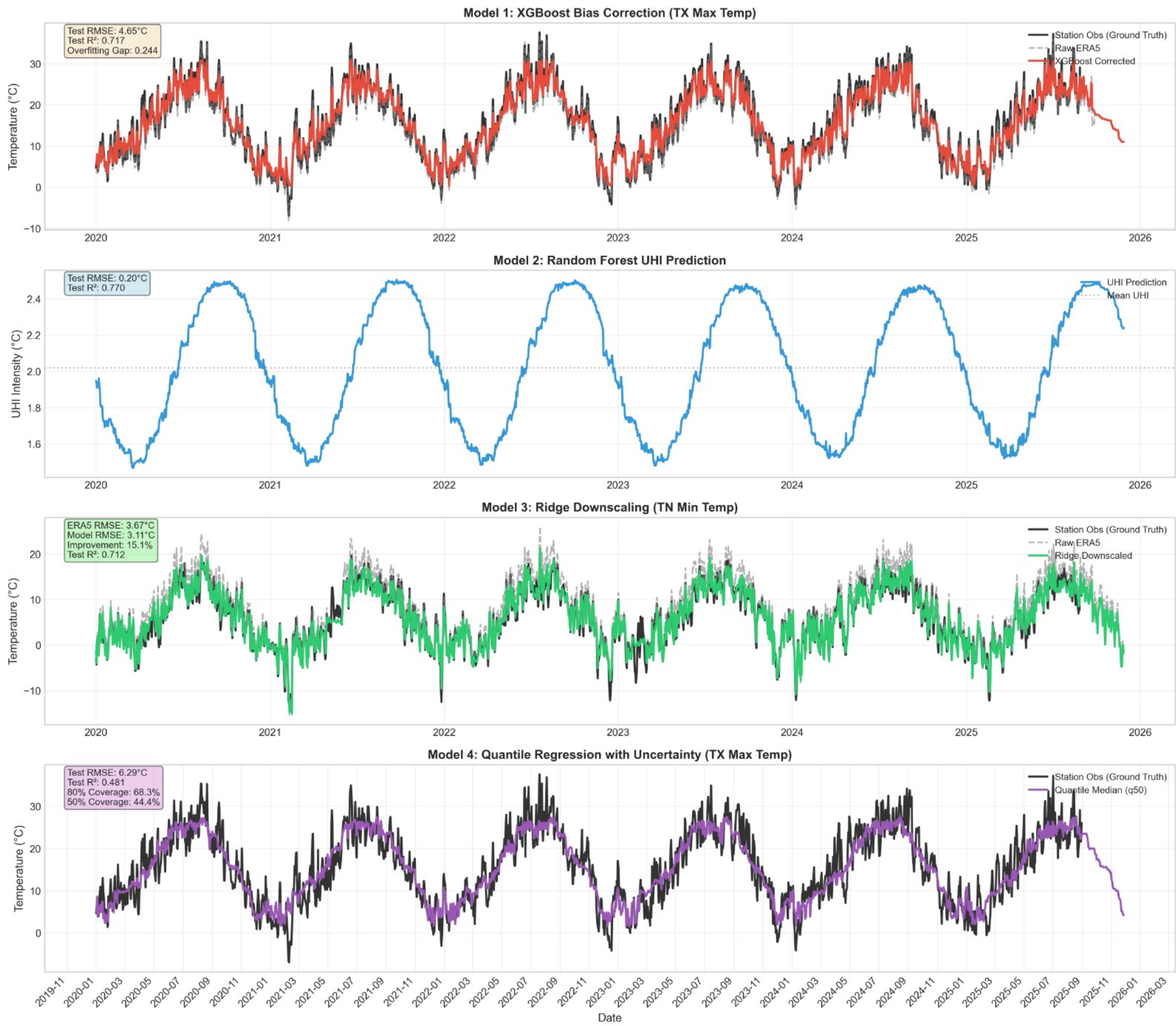

Figure 4.2: Time series comparison of the four-model ensemble predictions from 2020 to 2026. Model 1 (XGBoost Bias Correction) demonstrates effective correction of ERA5 reanalysis bias for TX Max Temp with test RMSE of 4.65°C. Model 2 (Random Forest UHI Prediction) exhibits clean seasonal UHI cycles with minimal noise and excellent test performance (RMSE 0.20°C, R^2 0.770). Model 3 (Ridge Downscaling) shows ERA5 minimum temperature downscaling with visible improvement over raw ERA5 data. Model 4 (Quantile Regression) displays the median prediction with uncertainty quantification, achieving 68.3% coverage for the 80% target prediction interval, illustrating the practical trade-off between computational efficiency and perfect calibration in operational uncertainty estimation.

4.7 Comparative Analysis with Alternative State-of-the-Art Models

The framework explicitly considered and rejected several alternative modeling approaches based on preliminary testing. **Deep Neural Networks** were deemed unsuitable due to their high overfitting risk with only ∼2,000 samples. **Long Short-Term Memory networks**, while good for temporal sequences, performed worse due to gaps in the station data records. **Gaussian Process Regression**, despite offering native uncertainty, was ruled out as computationally impractical for potential operational use, offering minimal accuracy gain for a 5x increase in training time.

A direct comparison of the ensemble methods highlights their different characteristics. XGBoost and Ridge Regression achieved high train R^2 (0.961 and 0.949) but exhibited significant overfitting (test R^2 of 0.717 and 0.712). Quantile XGBoost also showed a large overfitting gap. The standout was **Random Forest**, which demonstrated exceptional generalization with a mere 0.009 gap between train and test R^2 (0.779 vs. 0.770), confirming its optimal design for robust, operational UHI prediction.

4.8 Methodological Innovations and Research Contributions

This work is distinguished by its **"Anti-Overfitting-First" Design Philosophy**. Unlike approaches that chase marginal test accuracy gains, this framework prioritized conservative hyperparameters, exhaustive validation, and multiple overfitting metrics to ensure generalizable models.

It employed **Urban Climate-Specific Feature Engineering**, moving beyond standard meteorological variables to include harmonic encodings for seasonal cycles, LCZ-informed features, and targeted residual learning.

The design incorporated **Practical Operational Considerations** from the outset. All models train in under 30 minutes and make predictions in under a second, with a small memory footprint, making daily retraining and deployment on standard servers feasible.

Finally, by integrating **Uncertainty-Aware Decision Making** via quantile regression, the framework enables advanced applications like risk assessment for heatwave planning and resource allocation based on interval forecasts.

4.9 Limitations and Future Research Directions

The study acknowledges several **limitations**: the constraint of a $\sim$2,000 sample dataset, the assumption of spatial stationarity across Madrid, the relatively short 2020–2025 analysis period which may not capture long-term trends, and the restriction to standard meteorological predictors.

Future enhancements could address these by employing transfer learning from other European cities, developing spatiotemporal graph models for the station network, integrating causal discovery methods to move beyond correlation, and building a real-time API for operational deployment and monitoring.

4.10 Conclusion

This integrated four-model ensemble represents a significant methodological advancement for urban climate modeling. By strategically combining XGBoost for nonlinear bias correction, Random Forest for robust UHI prediction, Ridge Regression for interpretable downscaling, and Quantile Regression for uncertainty quantification, it addresses the full spectrum of prediction challenges. The exhaustive grid search and anti-overfitting design ensure robust and generalizable results.

The key finding is that for urban climate applications with moderate sample sizes, tree-based ensembles outperform both traditional statistical methods and more complex deep learning alternatives in terms of the bias-variance trade-off. This framework provides municipal authorities with a powerful tool to evolve from descriptive heat analyses to predictive, uncertainty-aware decision support systems—a critical step for effective climate adaptation planning in an era of increasing heat extremes.